

Elevare: A Platform for Enhancing Public Speaking and Soft Skills

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Abstract— Public speaking is a fundamental skill in academic and professional environments; however, many individuals struggle with speech anxiety and lack access to personalized, objective coaching. Existing AI-based public speaking assistants typically focus on isolated communication modalities, such as speech or facial expressions, while overlooking contextual understanding and comprehensive performance assessment. To address these limitations, this paper presents Elevare, an AI-powered multimodal public speaking coach that integrates emotion recognition, presentation comprehension, intelligent question generation, answer evaluation, and personalized feedback within a unified framework. Elevare consists of three interconnected modules. The first employs deep learning models for multimodal emotion recognition, achieving 92.9% accuracy in Speech Emotion Recognition using a Stacked BiLSTM, 84.46% accuracy in Facial Emotion Recognition using an optimized GiMeFive architecture, and 89.5% accuracy in body language emotion recognition using a Temporal Convolutional Network (TCN). The second module enhances content mastery by generating context-aware practice questions from presentation slides using a Semantic Graph Conversational Question Generation (SG-CQG) pipeline, while a zero-shot Retrieval-Augmented Generation (RAG) framework powered by Llama 3.3-70B evaluates user responses with a grading accuracy of 72.48%. Finally, a Dual-Agent Feedback Architecture based on Gemini 2.5 Flash combines objective performance analysis with empathetic coaching to produce actionable and personalized feedback. Experimental results demonstrate that Elevare provides a comprehensive, interactive, and content-aware coaching experience, enabling users to improve both presentation delivery and subject mastery while advancing AI-assisted public speaking training.

Keywords—Artificial Intelligence, Temporal CNN, Large Language Models, Public Speaking Anxiety

I. INTRODUCTION

The Effective public speaking is a fundamental competency in both academic and professional environments. Although technical knowledge, innovative ideas, and domain

expertise are essential for success, their impact is significantly diminished if they cannot be communicated with clarity, confidence, and persuasion [1]. Successful presentations depend not only on verbal communication but also on nonverbal behaviours, including facial expressions, vocal characteristics, and body language, which collectively influence audience engagement, speaker credibility, and message effectiveness. However, glossophobia—the fear of public speaking—remains one of the most prevalent forms of social anxiety, substantially impairing an individual's ability to communicate effectively.

Recent studies highlight the widespread nature of this problem among university students and early-career professionals. Approximately 50% of university students experience high levels of presentation anxiety, while an additional 42% report moderate anxiety during public speaking situations [2,3]. Such anxiety often leads to presentation avoidance, reduced participation in collaborative activities, diminished academic performance, and missed career opportunities. If left untreated, these difficulties may further develop into persistent social anxiety that negatively affects both personal and professional development [4,5]. Beyond academia, ineffective communication has become a major barrier to employability, as employers increasingly regard communication, confidence, and interpersonal skills as essential indicators of future workplace performance. Consequently, many qualified candidates fail to secure employment despite possessing strong technical competencies, emphasizing the growing demand for accessible and effective public speaking training solutions [6, 7].

Recent advances in artificial intelligence, affective computing, and large language models have accelerated the development of automated presentation assessment systems. Nevertheless, existing approaches remain limited in both scope and functionality. Most current solutions rely on unimodal analysis, providing feedback based solely on isolated metrics such as speech rate, filler words, or facial

expressions without considering the interaction among multiple communication channels or the semantic context of the presentation itself [8,9]. Furthermore, these systems rarely incorporate content-aware evaluation, generating generic practice questions instead of questions derived directly from the speaker's presentation material. As a result, they fail to simulate realistic presentation scenarios or detect inconsistencies between verbal and nonverbal communication, such as confident speech accompanied by anxious facial expressions or body posture.

To overcome these limitations, this paper presents Elevare, a comprehensive AI-powered multimodal public speaking coach that integrates emotion recognition, contextual presentation understanding, conversational question generation, answer evaluation, and personalized coaching into a unified intelligent framework. The proposed system simultaneously analyses speech, facial expressions, and body language through dedicated deep learning models to provide a holistic assessment of presentation performance. In addition, Elevare automatically generates context-aware questions from user-uploaded presentation slides using a Semantic Graph Conversational Question Generation (SG-CQG) pipeline and evaluates user responses through a zero-shot Retrieval-Augmented Generation (RAG) framework. To ensure reliable and personalized coaching, the system further employs a dual-agent large language model architecture that separates objective analytical reasoning from empathetic feedback generation, producing comprehensive and actionable recommendations for performance improvement.

The main contributions of this work can be summarized as follows. First, we propose a synchronized multimodal emotion recognition framework that integrates speech, facial, and body-language analysis using specialized deep learning architectures. Second, we introduce an emotion incongruence detection mechanism capable of identifying inconsistencies across communication modalities to reveal hidden confidence gaps. Third, we develop a content-aware presentation defense environment that automatically generates contextual questions and evaluates user responses using retrieval-augmented large language models. Finally, we design a dual-agent feedback architecture that combines objective performance assessment with psychologically informed coaching to deliver personalized, explainable, and actionable recommendations for improving public speaking performance.

The remainder of this paper is organized as follows. Section II reviews related work in multimodal emotion recognition and AI-assisted presentation coaching. Section III presents the proposed methodology and system architecture. Section IV discusses the experimental setup and performance evaluation, while Section V concludes the paper and outlines future research directions.

II. RELATED WORK

Reviewing and analyzing relevant academic literature and existing solutions offers valuable insights into the methodologies, benchmark datasets, and performance standards established in the field of automated communication analytics. Prior to detailing our specific implementation, we conducted an extensive review of multiple surveys and implementation papers covering the broader landscape of human behavior analysis and conversational AI. To maintain

a concise and targeted overview, the studies highlighted in the following subsections represent the most critical and highly relevant advancements directly informing the core pipelines of the Elevare framework: body language evaluation, facial emotion recognition, speech emotion recognition, conversational question generation, and Question Answer validation and Feedback Generation [10 - 22].

Vaijayanthi Sekar and J. Arunnehr [10] developed a human emotion recognition workflow utilizing body postures from the GEMEP dataset, which features 145 video recordings across 17 emotion classes. Their data preprocessing pipeline involved video-to-frame extraction, automated skeleton pose detection to map body joints, and coordinate normalization between 0 and 1 to standardize spatial data. These refined kinematic features were then evaluated across several machine learning classifiers, including Support Vector Machine (SVM), Decision Tree, and Naïve Bayes. Ultimately, the K-Nearest Neighbors (KNN) model delivered the best results, achieving a peak classification accuracy of 97.1% and demonstrating the high discriminative power of normalized posture coordinates.

T. Wang et al. [11] introduced a full-body emotion recognition framework evaluated across five public datasets: EGBM, KDAE, Emilya, MPI, and DMCD. The pipeline began with extracting and normalizing 3D skeletal joint coordinates, followed by a specialized pseudo-energy-based scale selection algorithm to isolate the most emotionally informative temporal segments. These segments were processed through an architecture featuring a multiscale spatio-temporal convolutional neural network (CNN) with hierarchical fusion mechanisms, which integrated posture covariance matrices and 3D posture image representations to jointly learn spatial and temporal features. This framework yielded excellent results, achieving average classification accuracies exceeding 86% across all evaluated datasets and consistently outperforming state-of-the-art methods.

Bingyu Nan et al. [12] developed a visual self-attention mechanism network for facial expression recognition using the RAF-DB dataset, which contains 30,000 facial images distributed across 7 emotion classes. Their data preprocessing pipeline involved automated face detection via RetinaFace, precise face alignment, and data augmentation through random horizontal flipping. The processed images were then fed into an advanced architecture featuring a Truncated ConvNeXt network paired with a specialized Detail Extraction Block designed to isolate subtle facial micro-expressions. This framework yielded highly successful results, achieving a peak classification accuracy of 97% and demonstrating the effectiveness of self-attention mechanisms in capturing fine-grained emotional features.

J. Wang and L. Kawka [13] proposed GiMeFive, a lightweight and interpretable model tailored for facial emotion classification using aligned and cropped images from the RAF-DB dataset, encompassing roughly 10,000 images across 6 primary emotion classes. Their pipeline kept preprocessing minimal, relying strictly on resizing images to a 64×64-pixel resolution and standard normalization without applying random spatial augmentations. The underlying architecture featured a compact convolutional design equipped with batch normalization and dropout, trained using standard cross-entropy loss with stochastic gradient descent. This streamlined workflow delivered efficient results,

achieving a baseline test accuracy of 86.5% while maintaining high computational simplicity.

Shahana Yasmin et al. [14] introduced a speech emotion detection framework evaluated on a massive combined dataset of 12,424 audio-visual samples from RAVDESS, TESS, SAVEE, EmoDB, and CREMA-D across 7 emotion classes. The data preprocessing pipeline consisted of removing silence segments longer than 200ms, resampling the remaining audio to 22,050 Hz, and applying data augmentations including Gaussian noise injection, pitch shifting, and time stretching. These processed signals were then passed into a hybrid architecture combining a Deep Conditional Random Field stacked with a Bidirectional LSTM (DCRF-BiLSTM). This configuration yielded highly robust results, achieving a final classification accuracy of 94.04% by effectively capturing complex acoustic and sequential temporal properties.

Akinpelu, Viriri, and Adegun [15] proposed an enhanced Speech Emotion Recognition (SER) system using the TESS and EMO-DB benchmark datasets. Their engineering pipeline involved applying noise reduction, signal normalization, and mel-spectrogram extraction to effectively transform raw 1D audio waves into 2D image-like visual representations. These spectrogram inputs were then processed using a lightweight Vision Transformer (ViT) architecture that leverages self-attention mechanisms to capture fine-grained spatial and emotional patterns. The experimental results demonstrated exceptionally strong performance, achieving specific accuracies of 98% on TESS, 91% on EMO-DB, and 93% on the combined datasets, underscoring the efficiency of transformer-based visual modeling for vocal audio analysis.

Xuan Long Do et al. [16] proposed SG-CQG, a two-stage framework for answer-unaware Conversational Question Generation (CQG) evaluated on the CoQA dataset, which contains 127,000 question-answer pairs. Their operational pipeline maps out the generation process by first utilizing a "what-to-ask" module that constructs a semantic graph from the text context to select key rationale sentences non-sequentially. Next, a "how-to-ask" module generates the final questions guided by explicit question type control signals and passes them through filtering heuristics to eliminate low-quality outputs. This framework achieved state-of-the-art results on CoQA, securing a BERTScore-entailment of 81.99, a Conv-Distinct score of 62.27, and a Context Coverage of 68.52%, outperforming baselines like T5, BART, and GPT-2 while human evaluations confirmed its superior factuality and conversational alignment.

Subhankar Maity, Aniket Deroy, and Sudeshna Sarkar [17] introduced an automated question generation framework designed for educational domains using the EduProbe dataset, which comprises 3,502 question-answer pairs from school textbook passages. Their engineering pipeline explores three distinct setups: In-Context Learning (ICL), Retrieval-Augmented Generation (RAG), and a novel Hybrid Model that combines retrieval-based context enrichment with few-shot prompt configurations. The experimental results showed that while standalone ICL (with $k=7$ examples) achieved the highest automated text scores with a ROUGE-L of 55.95 and a BERTScore of 75.92, the Hybrid Model (with $k=5$ examples and $m=5$ retrieved contexts) won the human evaluations. The hybrid approach recorded superior educational marks of 4.84 for grammaticality, 4.74 for appropriateness, 4.25 for relevance, and 4.02 for complexity.

Menna Fateen et al. [18] implemented a modular retrieval-augmented generation (RAG) framework for automatic short answer scoring and feedback generation using the `saf_communication_networks_english` dataset, which contains 300 technical communication network responses. Their operational pipeline utilized a zero-shot learning approach engineered via DSPy ChainOfThought prompting built onto a Llama 3.3-70b model backbone to dynamically compute strict classification labels and generate actionable feedback text. This automated setup yielded highly precise results, achieving a peak scoring accuracy of 72.48% and successfully demonstrating the system's capacity to deliver objective, explainable coaching metrics under entirely zero-shot conditions.

Thuy Vu and Alessandro Moschitti [19] introduced AVA, an automatic evaluation framework designed to assess the accuracy of question answering systems using over two million evaluation triplets compiled from the WikiQA, Natural Questions, and TREC-QA benchmarks. Their engineering pipeline involved transformer-based textual tokenization followed by sequential joint encoding of the question, reference answer, and system candidate response to lock in crucial contextual boundaries. These vectors were then processed through deep transformer language models that mapped high-dimensional sentence embeddings to compute final semantic similarity scores. The framework delivered highly reliable results, achieving an F1 score of 74.7% in predicting human grading decisions and an exceptionally low Root Mean Square Error (RMSE) between 0.02 and 0.09.

Soham Padia et al. [20] proposed a multimodal AI-driven framework for enhancing public speaking skills using a custom dataset of nearly 800 public speaking videos. Their processing pipeline involved video frame extraction, spatial resizing, and automated speech transcription to enable concurrent visual and textual analysis. The system's architecture integrated rule-based algorithms with multimodal tracking to dynamically extract verbal metrics—such as speaking rate, pitch modulation, and filler word frequency—alongside non-verbal facial expressions. By synthesizing these cross-channel metrics into structured graphical analyses, the framework successfully generated detailed, action-oriented feedback reports that accurately map out a user's behavioral strengths and targeted areas for physical delivery improvement.

N. Basta et al. [21] presented DIALEVAL, a type-theoretic dual-agent framework designed to automate the evaluation of instruction adherence and dialogue coherence. To eliminate systematic errors common in uniform LLM scoring, their pipeline splits the task between two specialized Claude-3.5-Sonnet agents: the first agent is strictly dedicated to extracting and decomposing dialogue text into typed predicates, while the second independent agent runs type-specific satisfaction assessments based on human-aligned criteria. Evaluated across multiple LLM backbones, this dual-agent architecture delivered highly precise results, achieving a 90.38% evaluation accuracy and a substantially stronger alignment with human judgment than state-of-the-art single-agent methods.

Lucas Jasper Jacobsen and Kira Elena Weber [22] investigated the efficacy of large language models as automated feedback providers in higher education, specifically focusing on teacher training. Their methodology

pipeline involved creating a theory-driven prompt engineering manual and constructing three distinct prompt structures that varied in pedagogical quality to guide ChatGPT-4 as the core generative engine. The study's results demonstrated that when guided by the highest-quality structured prompt configuration, the LLM not only outperformed novice human feedback but also surpassed domain experts in crucial metrics including explanation clarity, targeted question generation, and diagnostic specificity, validating precisely engineered LLMs as highly efficient alternatives to traditional human feedback mechanisms.

A comprehensive review of the literature reveals that existing AI-based public speaking systems primarily address isolated tasks rather than providing an integrated coaching framework. Current approaches independently analyze speech, facial expressions, or body language, while lacking synchronized multimodal assessment, content-aware question generation from presentation materials, and rigorous answer validation. Moreover, existing feedback mechanisms commonly rely on single-agent language models, leading to generic or biased assessments. To overcome these limitations, the proposed Elevare framework introduces a unified, context-aware architecture that integrates multimodal emotion recognition, presentation-specific Q&A evaluation, and a Dual-Agent feedback mechanism to deliver objective and personalized coaching.

III. METHODOLOGY

The Elevare system follows a three-tier architecture comprising a Presentation Layer, an Application Layer, and a Database Layer, as illustrated in Fig. 1. User inputs — consisting of presentation slides, a recorded presentation video, and spoken responses to generated questions — enter through the Presentation Layer and are routed to the Application Layer for processing.

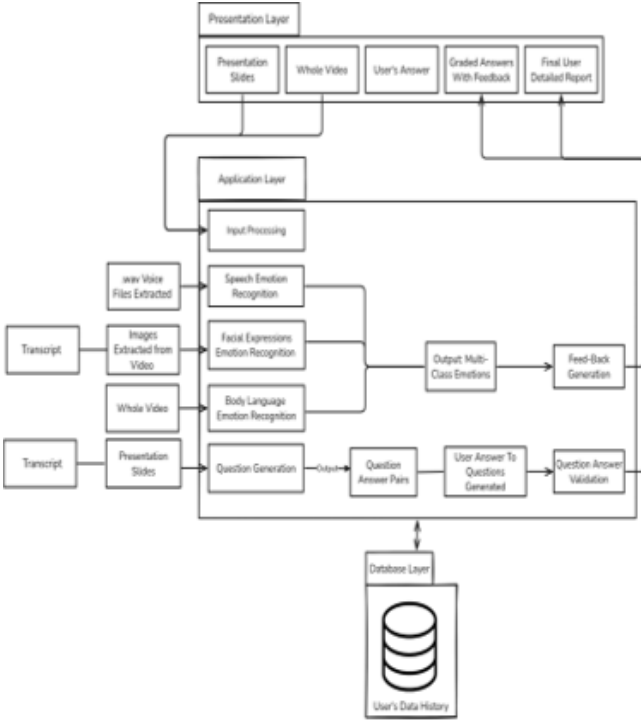


Fig. 1. System architecture of Elevare.

The Application Layer houses five parallel processing modules: Speech Emotion Recognition, Facial Emotion Recognition, Body Language Emotion Recognition, Question Generation, and Feedback Generation. The three emotion recognition modules operate concurrently on their respective input streams — audio, video frames, and skeletal data — producing a unified set of multi-class emotion outputs that are forwarded to the Feedback Generation module. In parallel, the Question Generation module processes the presentation slides independently to produce question-answer pairs, which are subsequently presented to the user for response before being passed to the Question Answer Validation module. The Database Layer persists user account data and historical performance reports across sessions.

The system follows two sequential processing paths that converge at the Feedback Generation module. The first path processes the presentation video through the three emotion recognition modules in parallel, extracting affective signals across speech, facial, and body modalities simultaneously. The second path processes the presentation slides through the Question Generation pipeline, producing contextually grounded questions that simulate an audience Q&A session, with user responses evaluated by the Answer Validation module. The outputs of both paths — multimodal emotion metrics, acoustic and semantic features, and answer validation scores — are consolidated by the Feedback Generation module, which synthesizes them into a structured, personalized performance report delivered to the user through the Presentation Layer.

A. Facial Emotion Recognition Module

1. ConvNeXt + Detail Extraction Block Model

Input images were resized to 224×224 pixels, expanded to three channels, and normalized using ImageNet mean and standard deviation values to ensure compatibility with the pretrained backbone.

Feature extraction is built on a truncated ConvNeXt [23] backbone that captures low- and mid-level facial representations. ConvNeXt is originally divided into four stages of progressively deeper semantic representations; however, only the first two stages are retained, as the higher semantic stages introduce unnecessary complexity and risk overfitting given the dataset's relatively small size. This truncation yields a lightweight backbone focused on texture-level features relevant to facial expression analysis [24]. To compensate for the reduced backbone depth, a custom Detail Extraction Block (DEB) is placed after the truncated ConvNeXt to enhance sensitivity to subtle expression variations. The DEB applies a sequence of depthwise separable convolutions [25] with progressively smaller kernels, followed by spatial dropout and spatial pooling to aggregate information across the feature map. A self-attention mechanism then weights the most expressive facial regions, producing a compact yet discriminative feature representation passed to the classification head.

During training, augmentation techniques including random horizontal flipping, rotation, color jitter, and random erasing were applied to improve robustness against variability in pose, illumination, and occlusion. Techniques involving heavy geometric distortion and region-level cropping were evaluated but excluded, as they introduced unrealistic distortions that degraded generalization [26,27]. To address

class imbalance, minority classes were oversampled through augmentation, class-weighted loss was applied, and Label-Distribution-Aware Margin (LDAM) loss with Deferred Re-Weighting (DRW) [28] was adopted to improve decision boundary quality for rare emotion classes, following a two-stage warm-up strategy in which standard cross-entropy was first applied with the backbone frozen before enabling LDAM+DRW in the second stage.



Fig. 2. ConvNeXt + Detail Extraction Block Architecture based on [24].

The end-to-end architecture of the proposed model is illustrated in Fig. 2. The pipeline begins with the truncated ConvNeXt backbone, which extracts low- and mid-level spatial features from the input image. These features are then passed through the Detail Extraction Block, which refines them using multi-scale depthwise convolutions, spatial dropout, and a self-attention mechanism. The refined feature maps are aggregated via global pooling and mapped to the eight emotion categories through a fully connected classification head. The model was trained using the AdamW optimizer [11] with a weight decay of 1×10^{-4} , with mixed precision training and gradient clipping applied to improve stability. A differential learning rate strategy was adopted, assigning a learning rate of 1×10^{-5} to the backbone and 1×10^{-4} to the DEB and classification head, with early stopping applied based on balanced accuracy to prevent overfitting.

2. GiMiFive Model

Input images were resized to 64×64 pixels and normalized using standard mean and standard deviation values consistent with the GiMeFive architecture's input requirements [29].

Given the significant class imbalance present in the training data, a class-aware augmentation strategy was adopted rather than applying uniform augmentation across all classes. Minority classes — specifically Anger, Disgust, and Fear — were heavily augmented to increase their representation, while majority classes received only light augmentation to preserve their well-represented feature distributions. Mixup augmentation [30] was additionally applied at $\alpha=0.2$, interpolating between pairs of training images and their labels to encourage the model to learn smoother decision boundaries between emotion classes. To further address class imbalance, Deferred Re-Weighting (DRW) was adopted, training the model under standard cross-entropy loss for an initial period before introducing square-root-scaled class weights in later epochs, allowing the model

to first establish stable feature representations before shifting focus toward minority classes.

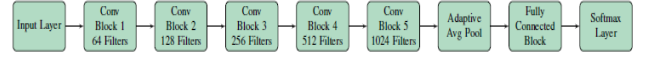


Fig. 3. GiMeFive Model Architecture based on [29].

The GiMeFive model [29] is a custom 15-layer convolutional neural network that takes a 3-channel 64×64 image as input and produces class probabilities through five convolutional blocks followed by a fully connected classification head. Each convolutional block consists of a 3×3 convolution with padding, Batch Normalization, ReLU activation, 2×2 Max Pooling, and a Dropout layer at 20%, with filter depth progressively increasing from 64 to 1,024 channels across the five blocks. Following the convolutional blocks, an Adaptive Average Pooling layer aggregates spatial information before three fully connected layers map the learned representations to the output space, terminating in a Softmax classification layer. With approximately 10.5 million parameters, the architecture achieves a strong balance between efficiency and classification performance, being significantly more compact than VGG-16 and ResNet-34 while remaining competitive on facial expression benchmarks. The model was trained using the SGD optimizer with a learning rate of 0.001, momentum of 0.9, and weight decay of 1×10^{-4} , over 130 epochs with a warm restart learning rate schedule.

B. Speech Emotion Recognition Module

The collected speech recordings underwent several preprocessing steps to improve signal quality and ensure consistency across the combined dataset. Silence removal was first applied, where leading or trailing silence exceeding 200 ms was trimmed by approximately 70% to emphasize emotionally relevant speech content and reduce non-informative regions. All audio clips were then resampled to a unified sampling rate of 22,050 Hz to maintain consistency across the different dataset sources. Waveform structures were further standardized after cleaning to ensure consistent signal characteristics during feature extraction.

To improve model robustness and reduce overfitting, multiple audio augmentation techniques were applied during training. Gaussian noise injection was performed at noise levels of 0.035, 0.025, and 0.015 to simulate real-world recording conditions and environmental noise. Pitch shifting was applied in three stages using rate parameters of 0.70, 0.80, and 0.70 to introduce variation in vocal tone without altering audio duration. Time stretching was additionally applied with speed factors of 0.8, 0.9, and 0.7 to simulate different speaking rates while preserving pitch information. These techniques increased dataset diversity, helped address class imbalance, and improved generalization across different speakers and acoustic environments.

Feature extraction was performed using the Librosa audio analysis library to convert raw speech signals into meaningful representations suitable for model training. A total of 190 features were extracted from each audio sample, grouped into six categories: Mel-Frequency Cepstral Coefficients (MFCCs), capturing vocal tract characteristics and emotional variations, comprising 20 MFCC coefficients alongside their delta, delta-delta, and standard deviation counterparts; Chroma features, capturing tonal and harmonic characteristics

across 12 pitch classes via three variants — Chroma-STFT, Chroma-CQT, and Chroma-CENS — yielding 36 features; Log-Mel Spectrogram (LMS) features, providing a time-frequency representation that preserves emotional speech patterns, yielding 64 features; Spectral Contrast, measuring the difference between spectral peaks and valleys to identify harmonic and non-harmonic speech components, yielding 6 features; Root Mean Square Energy (RMSE), reflecting speech intensity and loudness; and Zero Crossing Rate (ZCR), capturing voiced versus unvoiced speech transitions.

Following feature extraction, all features were normalized using Z-score standardization to achieve zero mean and unit variance. Principal Component Analysis (PCA) [31] was then applied to reduce feature dimensionality while preserving the most significant information from the extracted feature set, minimizing redundancy and reducing computational complexity. The optimal PCA configuration was determined experimentally at 140 components. The processed features were subsequently split into training and testing sets using an 80%–20% ratio to ensure reliable performance assessment and evaluate the generalization capability of the model.

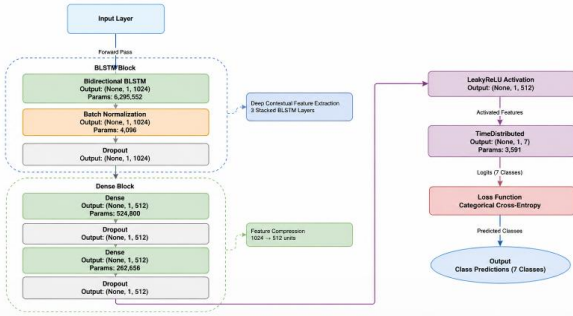


Fig. 4. BiLSTM Architecture adapted from [17].

The model receives the PCA-reduced feature set as input, encompassing representations derived from MFCCs, Chroma features, Log-Mel Spectrograms, Spectral Contrast, RMS Energy, and ZCR. The core component of the architecture is the Bidirectional Long Short-Term Memory (BiLSTM) layer [32], which learns temporal dependencies in speech sequences from both forward and backward directions, enabling the model to capture contextual emotional patterns occurring before and after each speech frame. A Dropout layer follows the BiLSTM layer to reduce overfitting by randomly deactivating neurons during training, after which a Dense layer transforms the learned temporal representations into higher-level emotional embeddings. A Softmax output layer then produces a probability distribution across the target emotion classes. The original architecture included a Deep Conditional Random Field (DeepCRF) layer to model sequential dependencies between output labels; however, this was removed as the single time-step input structure limited its ability to learn meaningful transition dependencies, and experimental evaluation confirmed that the BiLSTM layers alone were sufficient for accurate emotion recognition without any degradation in performance. The model was trained using the Adam optimizer with a learning rate of 1×10^{-4} and categorical cross-entropy loss, with a batch size of 256 over 500 epochs and early stopping applied to prevent overfitting.

C. Body Language Emotion Recognition Module

All models in this module operate on OpenPose skeleton sequences extracted from the MEED dataset. Keypoints with

a confidence score below 0.55 were treated as missing, consistent with prior work on keypoint filtering for pose-based recognition. For sequential models, missing values were recovered through temporal interpolation across neighboring frames, while for frame-level models (SVM, KNN), low-confidence keypoints were excluded from feature computation. All joint coordinates were translated relative to the mid-hip joint to remove absolute position dependency, with the neck joint used as a fallback anchor [33,32]. Scale normalization was applied using the Euclidean distance between the left and right shoulder joints, ensuring invariance to subject size and camera distance, with a fallback scale computed from the median distance of valid key points from the center. Finally, all sequences were standardized to 97 frames — consistent with the MEED dataset structure with shorter sequences padded by repeating the last frame.

The shared preprocessing pipeline was applied as described above. To address class imbalance in the MEED dataset, the Synthetic Minority Oversampling Technique (SMOTE) was applied to the training set, generating synthetic samples for underrepresented emotion classes through feature-space interpolation.

Following Vijayanthi and Arunehru [35], six pairwise Euclidean distances were computed from the normalized upper-body skeleton to capture arm extension and limb proximity to the torso: nose–right wrist, nose–left wrist, right wrist–right hip, left wrist–left hip, right elbow–right hip, and left elbow–left hip.

SVM is a supervised learning algorithm that finds the optimal hyperplane maximizing the margin between classes in the feature space. For non-linearly separable data, the kernel trick maps input features into a higher-dimensional space where a linear decision boundary can be established. An RBF kernel was employed with regularization parameter $C = 1$ and kernel coefficient $\gamma = \text{scale}$, following the experimental setting of Vijayanthi and Arunehru for body posture-based emotion recognition.

KNN is a non-parametric, instance-based classifier that assigns a class label by majority vote among the k nearest neighbors in feature space using Euclidean distance. With $k = 5$, it serves as a distribution-free baseline alongside SVM, following the experimental framework of Vijayanthi and Arunehru, applied here to the MEED dataset.

The shared preprocessing pipeline was applied as described above. Additionally, rotation normalization was applied to remove orientation bias across recordings. A subject-discriminative joint subset was then selected, retaining only 11 upper-body joints (indices 0–9 and 12), as upper-body joints carry the most discriminative information for emotion recognition, reducing input dimensionality while preserving classification-relevant structure. Class imbalance was addressed through inverse-frequency class weights applied to the loss function during training.

Three feature channels were computed from the normalized joint subset per frame: raw joint positions (x, y coordinates), per-joint velocity (frame-to-frame displacement), and per-joint acceleration (second-order displacement).

The ST-TR model [36] takes the skeleton sequence as input and projects it to a latent space via a 1×1 convolutional

embedding followed by batch normalization and ReLU activation, as illustrated in Fig. 5. The embedded features are then processed through a dual-branch block operating in parallel: the spatial transformer applies self-attention across joints per frame, augmented with an adjacency bias and relative positional encoding to capture inter-joint relationships; the temporal transformer applies self-attention across frames per joint, combined with a TCN branch and residual connection to model motion dynamics. The output is aggregated via global average pooling and passed to a classifier head consisting of a dropout layer (rate = 0.5) followed by a linear projection to 5 class logits. The model was trained for 45 epochs with a learning rate of 5×10^{-4} on sequences of 97 frames.

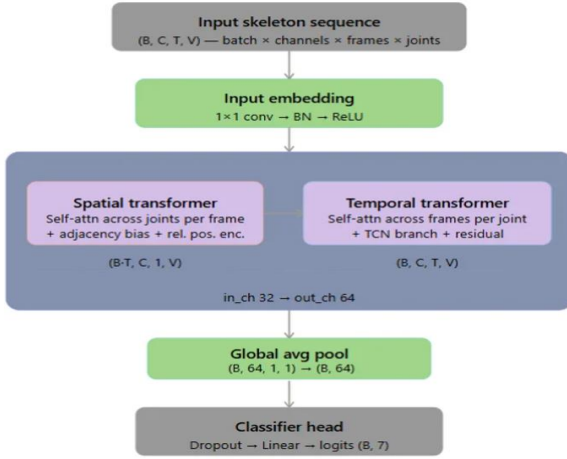


Fig. 5. Architecture of the Spatial-Temporal Transformer (ST-TR), adapted from [15].

The shared preprocessing pipeline was applied as described above. Class imbalance was addressed through inverse-frequency class weights applied to the loss function during training, assigning higher penalty to underrepresented emotion classes.

Following [37], five feature types were extracted per frame from the normalized skeleton sequence: pairwise Euclidean distances, joint angles, triangle areas formed by joint triplets, velocity, and acceleration. These features were concatenated per frame into a unified feature vector, which was then summarized across the temporal dimension using statistical aggregates — mean, standard deviation, median, and energy — over fixed chunks, yielding a flat vector as the final model input.

The proposed MLP consists of two fully-connected hidden layers of dimensions 512 and 256, each followed by a LeakyReLU activation and a dropout layer (rate = 0.338), as illustrated in Fig. 6. No batch normalization was applied. The output layer projects to 5 class logits, with the predicted class determined by argmax. The network was trained using AdamW with a learning rate of 0.00313 and weight decay of 1.78×10^{-5} . A ReduceLROnPlateau scheduler halved the learning rate upon 20 epochs without improvement, and early stopping was applied with a patience of 25 epochs. The loss function was weighted CrossEntropyLoss with inverse-frequency class weights.

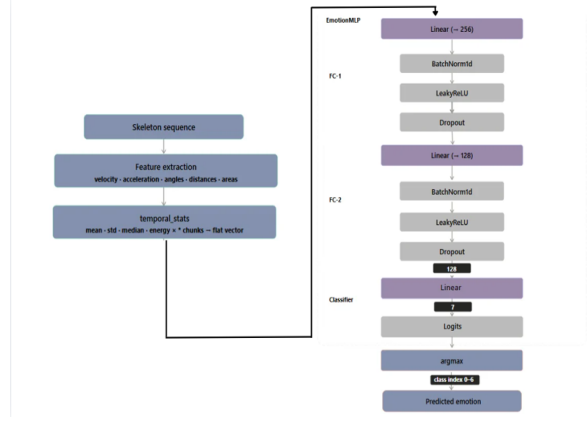


Fig. 6. Architecture of the proposed Multilayer Perceptron (MLP) for skeleton-based emotion recognition.

The shared preprocessing pipeline was applied as described above, with two additional model-specific steps. First, the 25 OpenPose BODY_25 joints were remapped to the 17-joint H36M format to standardize the skeleton representation; joints without a direct correspondence were approximated by averaging their OpenPose equivalents. Second, coordinates were normalized to the range $[-1, 1]$ using the estimated image dimensions (720×576). Class imbalance was addressed through data augmentation during training, including horizontal flipping with symmetric joint swapping, Gaussian noise injection on coordinates, and random temporal speed perturbation.

Three complementary feature streams were computed from the normalized skeleton sequence: joint positions (raw x, y coordinates), bone vectors (differences between connected joint pairs), and per-joint velocity (frame-to-frame displacement). Each stream captures a distinct aspect of motion — static pose, structural relationships, and temporal dynamics, respectively.

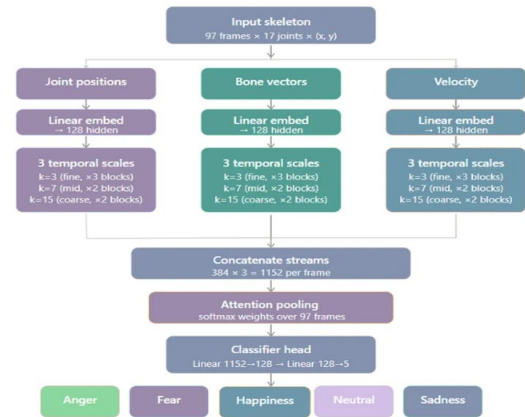


Fig. 7. Architecture of the proposed Temporal Convolutional Neural Network (TCN) for skeleton-based emotion recognition.

The proposed model processes the three feature streams in parallel through independent branches, as illustrated in Fig. 7. Each branch begins with a linear embedding layer projecting to 128 hidden dimensions, followed by multi-scale temporal convolutions at three kernel sizes: $k = 3$ (fine, $\times 3$ blocks), $k = 7$ (mid, $\times 2$ blocks), and $k = 15$ (coarse, $\times 2$ blocks), capturing short, medium, and long-range temporal dependencies simultaneously. The outputs of all branches are concatenated

per frame into a 1152-dimensional representation (384×3), which is then aggregated across the 97-frame sequence via attention pooling using softmax-weighted summation. The resulting vector is passed to a two-layer classifier head (Linear $1152 \rightarrow 128$, Linear $128 \rightarrow 5$) producing logits over the 5 emotion classes. The architecture and hyperparameters were determined through iterative experimentation.

D. Question Generation Module

Prior to question generation, the input PowerPoint presentations are processed through a pipeline that extracts and structures textual content into CoQA-format contexts. CoQA was selected as the target format since the question generation models SQ and CQG are pretrained on it, ensuring compatibility between the preprocessed input and the models' expected format.

Text is extracted from titles, body placeholders, and text boxes. Table content is converted into natural-language sentences by pairing column headers with their corresponding cell values. For image shapes, OCR is applied using Tesseract [36] to extract embedded text. Speaker notes are appended as additional context.

- **Text Cleaning:** Extracted text undergoes Unicode normalization and control character removal. URLs and lone slide numbers are discarded. Logical symbols are mapped to their textual equivalents, and bullet points are split into individual sentences.
- **Story Assembly:** Content from each slide is ordered as: title, body, image captions, and speaker note. Duplicate sentences are removed and all slides are concatenated into a single continuous passage.
- **Passage Chunking:** The passage is split into chunks of at most 350 tokens using NLTK sentence tokenization [37], preserving sentence boundaries. The 350-token limit was chosen to match the average context length in the CoQA dataset [35]. Each chunk is saved in CoQA format ready for input to the question generation stage.

The question generation module consists of two sequential stages: conversational question generation using the SG-CQG framework, followed by persona-based rephrasing guided by Bloom's Taxonomy [38].

1. Conversational Question Generation (SG-CQG)

Given an input context, the model constructs a Semantic Graph by identifying named entities and resolving coreferences, linking pronoun references back to their respective entities. This graph encodes the relational structure of the passage. A graph traversal algorithm then operates over this structure, incorporating the conversational history to select the most appropriate rationale — the sentence or concept that grounds the next question — ensuring that questions follow the logical flow of the document.

Once a rationale is selected, the how-to-ask module determines the form of the question. A question type classifier first produces an explicit control signal specifying the target question category — for instance, distinguishing between boolean yes/no questions and open-ended wh- span questions. This control signal, together with the selected rationale and the QG input, is passed to a fine-tuned T5 transformer [39] that synthesizes the final natural language question. The T5

generator ensures that the output is both grammatically well-formed and semantically grounded in the source context.

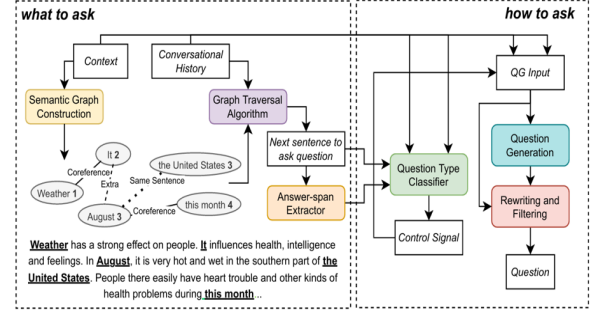


Fig. 8. SG-CQG model architecture, adapted.

2. Persona-Based Rephrasing via Bloom's Taxonomy

While Stage 1 produces factually grounded questions, real audiences approach the same material with different backgrounds, intentions, and cognitive goals. To capture this diversity, the QA pairs produced by SG-CQG are passed to a rephrasing stage that transforms them into persona-specific questions.

Bloom's Taxonomy [38] is a hierarchical framework for classifying educational objectives according to cognitive complexity. As shown in Table 1, it defines six levels in ascending order: Remember (L1), Understand (L2), Apply (L3), Analyze (L4), Evaluate (L5), and Create (L6), progressing from pure recall toward higher-order thinking.

Table 1. Bloom's Taxonomy cognitive levels [49].

Bloom's level	Description
Remember	Retrieve relevant knowledge from long-term memory.
Understand	Construct meaning from instructional messages, including oral, written, and graphic communication.
Apply	Carry out or use a procedure in a given situation.
Analyse	Break material into foundational parts and determine how parts relate to one another and the overall structure or purpose.
Evaluate	Make judgements based on criteria and standards.
Create	Put elements together to form a coherent whole; reorganise into a new pattern or structure.

Each persona is mapped to one of the six Bloom levels, and a structured prompt is constructed specifying the characteristics and objectives of each level, along with six few-shot examples — one per level — to guide the model's output style. This prompt, together with the original QA pairs and the source context, is submitted to LLaMA 3.3 70B Instruct, which rephrases each question to reflect the intent of the corresponding persona, as illustrated in Fig. 9.

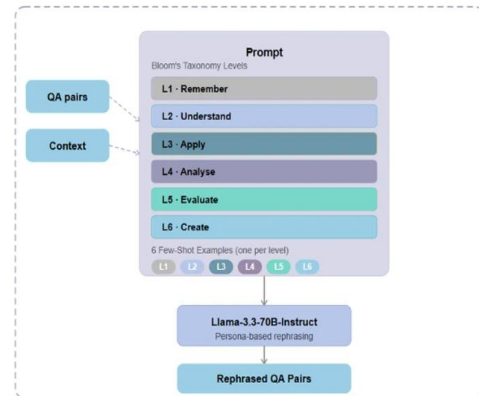


Fig. 9. Bloom's Taxonomy persona-based rephrasing module.

E. Question Answer Validation Module

Four techniques were considered for automatic short answer scoring with feedback: zero-shot Large Language Model (LLM) prompting (ASAS-F-Z), automatic few-shot prompt optimization (ASAS-F-Opt), Retrieval-Augmented Generation (RAG)-based few-shot scoring (ASAS-F-RAG), and a similarity-based majority-vote baseline. Since our use case involves open-ended, unpredictable presentation questions with no available labeled examples, independent evaluation of these approaches was conducted to determine the most suitable method for our deployment context. Zero-shot prompting generalizes best to new and unseen questions, making it the most appropriate choice given that question structure is variable and no retrieval database of labeled examples is available. Few-shot RAG, while producing more consistent outputs, requires a maintained database of labeled examples and a retrieval pipeline unavailable in our context [40]. Furthermore, since presentation questions are open-ended and highly variable, retrieved examples from a fixed database would rarely be semantically relevant enough to guide the model correctly. Accordingly, the zero-shot ASAS-F-Z approach was adopted as the final system.

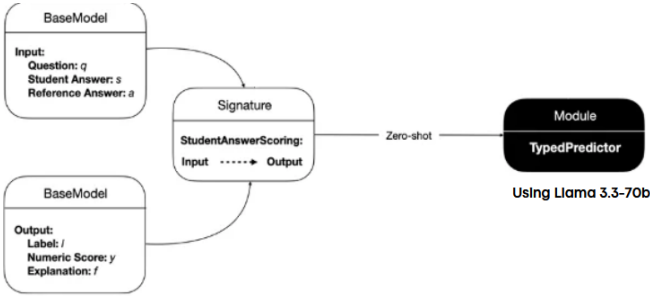


Fig. 10. High-level design of the Zero-Shot ASAS-F-Z module using DSPy.

The system is built using the DSPy (Declarative Self-improving Python) framework, which automates prompt generation and refinement by allowing developers to define input/output signatures rather than hand-crafting prompts, making the system more modular and adaptable to different question types. The backbone language model is Llama 3.3-70b, accessed via the Groq API. The module is defined through a DSPy Signature named `StudentAnswerScoring`, which specifies the input fields question, student answer, and reference answer and the expected output fields — a label, a numeric score, and an explanation. This signature is passed to a `TypedPredictor` module, which enforces strict type constraints on the model output and invokes the model in a zero-shot manner without any few-shot examples or fine-tuning [41].

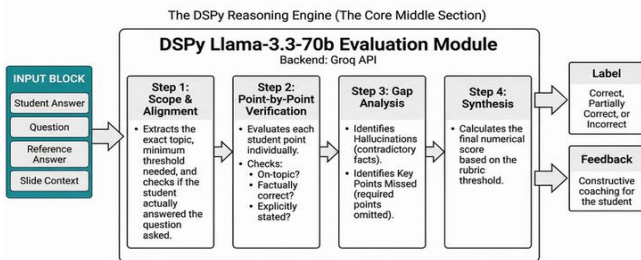


Fig. 11. Internal reasoning pipeline of the DSPy Llama 3.3-70b Evaluation Module

The internal reasoning pipeline consists of four sequential steps: Scope and Alignment, where the model extracts the question topic and verifies the student answer addresses it; Point-by-Point Verification, where each student point is evaluated for topical relevance and factual correctness; Gap Analysis, where hallucinations, contradictory facts, and omitted key points are identified; and Synthesis, where a final numeric score and a Label (Correct, Partially Correct, or Incorrect) are produced alongside constructive feedback. This multi-step structure reduces hallucinations, improves scoring consistency, and ensures each student point is evaluated independently before a final score is synthesized. Llama 3.3-70b was selected as the backbone model as it achieved the best Root Mean Square Error (RMSE) across the evaluated zero-shot models, indicating more stable and reliable numeric scoring on unseen questions, while also being fully available at no cost within the deployment environment.

F. Feedback Generation Module

The feedback pipeline comprises five components: Multimodal Inputs, Feature Extraction, a Reasoning LLM, a Coaching LLM, and Final Feedback output.



Fig. 12. Feedback Pipeline.

Raw audio and video are transformed into structured metrics during Feature Extraction, which are subsequently passed to the Reasoning LLM for objective threshold-based analysis, and finally to the Coaching LLM for translation into constructive, human-readable feedback [60].

1. Multimodal Feature Extraction

Raw audio is transcribed using faster-whisper, a CTranslate2-based reimplementation of OpenAI's Whisper, producing word-level timestamps, text segments, and duration metrics. faster-whisper was selected over the original Whisper implementation for its significantly reduced inference time and memory footprint at comparable transcription accuracy. Features extracted in this stage were selected based on established findings in communication science linking specific verbal and vocal behaviors to perceived speaker quality. Features are grouped into four categories: Speech Features, including words per minute, filler word detection, pause analysis, and stutter recovery detection; Emotion Analysis from speech, facial, and body language modules; Semantic and Structure features including sentence embeddings via SBERT, cosine similarity, coherence approximation, and topic drift detection; and Temporal Analysis covering silence gap detection. Frame-level acoustic features — loudness, pitch variation, and Signal-to-Noise Ratio per frame — are additionally extracted. The output of Stage 1 is a structured metrics.json file passed to Stage 2.

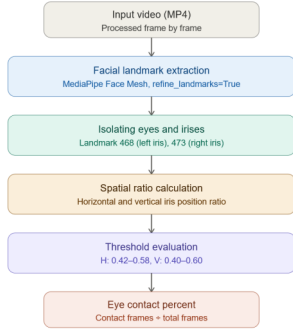


Fig. 13. Eye Gaze Model Pipeline.

The Eye Gaze module computes an Eye Contact Percent metric across the video using OpenCV and MediaPipe Face Mesh. MediaPipe Face Mesh is instantiated with `refine_landmarks=True`, enabling high-precision iris localization via dedicated iris landmarks (468 and 473) without requiring a separate iris-tracking network. Gaze direction is determined through spatial iris-to-eye-boundary ratios rather than a learned classifier, offering three advantages: interpretability, as thresholds can be manually inspected and tuned; adjustability, as thresholds can be recalibrated for different camera conditions without retraining; and robustness to domain shift, as the method depends on geometric relationships rather than training-distribution-specific visual features. The pipeline processes each frame by extracting 468+ facial landmarks, computing the iris position within each eye's bounding box as horizontal and vertical ratios, and averaging across both eyes to account for facial angle. A frame is classified as maintaining eye contact if the horizontal ratio falls within 0.42–0.58 and the vertical ratio within 0.40–0.60. The Eye Contact Percent is the proportion of such frames across the full video.

2. Analytical Reasoning Engine

Stage 2 is a controlled, threshold-based analytical system designed to minimize subjective framing in its output. It ingests the `metrics.json` from Stage 1 alongside a reference JSON containing communication science thresholds for each metric. Gemini 2.5 Flash [43] was selected as the backbone for this stage, as the task of generating structured analytical output from pre-computed metrics does not require advanced reasoning or long-context capabilities, and its availability within the development environment supported iterative refinement without API costs. The agent assesses each metric relative to its threshold, ranks deviations by severity, identifies the top three critical issues, and produces a structured JSON output consisting of issue names with severity and category, per-metric observations with status flags, impact descriptions, suggestion phrases, and a ranking rationale field. It is noted that large language models carry inherent biases from training data; this design constrains output behavior at the system level rather than eliminating model-level bias.

Beyond threshold-based analysis, Stage 2 detects cross-modal behavioral patterns through three dedicated algorithmic components. Emotion Incongruence Detection flags mismatches between positive facial expressions and negative vocal tone, grounded in research on emotional masking and cross-modal conflict detection. Confidence Gap Detection flags positive speech content paired with tense body posture, grounded in research linking closed posture to speaking

anxiety [75]. Discourse Structure Modeling identifies discourse breaks from sustained drops in sentence-level semantic similarity or sharp topic shifts computed in Stage 1. These components compute signal mismatches algorithmically before passing results to the LLM, rather than relying on the model to infer them from raw inputs.

3. Coaching and Feedback Agent

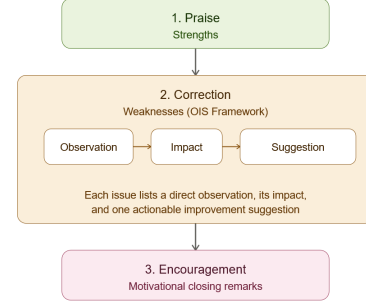


Fig. 14. Report Structure Diagram based on Sandwich Method [63]

Stage 3 maps the structured JSON output of Stage 2 into human-readable feedback following three enforced structural rules. The Sandwich Method is applied, opening with genuine praise, followed by constructive OIS breakdowns, and closing with motivational encouragement. OIS Delivery Execution ensures each issue is paired with a data-grounded observation, its audience impact, and one actionable improvement phrase. Strict output boundaries are enforced: only the top three issues are addressed, and the agent is explicitly constrained from contradicting any data produced by Stage 2. The final output is a JSON structure containing a `praise` field, a `constructive_feedback` array of OIS narratives, an `encouragement` field, and a `full_formatted_text` compilation field.

Multimodal emotion signals — facial, speech, and body are integrated into the pipeline at two stages [42]. In Stage 2, they are passed as raw analytical inputs alongside metric data for objective assessment. In Stage 3, they inform the encouragement section and closing remarks, grounding feedback in the speaker's actual emotional delivery. Three integration mechanisms are applied. Emotional Congruency Scores assess cross-modal consistency between facial, body, and speech emotions, flagging mismatches as Delivery Incongruency metrics. Confidence and Anxiety Indices aggregate discrete emotions into composite behavioral states following the dimensional approach in affective science. Emotional Trajectory tracks emotional shifts across the presentation timeline, informed by the serial position effect, treating a stable baseline with strong opening and closing engagement as a favorable pattern and flagging deviation spikes as potential indicators of reduced composure.

IV. EXPERIMENTAL RESULTS

This section presents the performance evaluation of the Elevare system. To thoroughly assess the architecture's capability as a multimodal public speaking coach, the system was evaluated across its five primary modules: facial emotion, body emotion, speech emotion, question generation, question answer validation.

A. Datasets

Facial Emotion Recognition Dataset: The Dataset FERplus was used. This dataset has ~35k images with 8 emotion classes. For the preprocessing, refer to the table for each trial. For the measurement metrics, accuracy, macro F1 and macro F1 recall were used.

Speech Emotion Recognition Dataset: The Dataset that was used, RTSEC, is a combination of the following dataset: RAVDESS, TESS, SAVEE, EmoDB, CREMA-D. For the measurement metrics, PCA, accuracy, and macro f1 were used.

Body Language Emotion Recognition Dataset: The dataset MEED was used. This dataset has ~4k recordings with 7 emotion classes. The following preprocessing was used: Key point Interpolation, Centering, Scale Normalization, Rotation Normalization, Sequence Length Standardization, Joint Subset Selection, Class Balancing. Accuracy was used as measurement.

Question Generation Dataset: The dataset CoQA was used. This dataset has ~127k Q&A pairs. QG dataset was also used during the trials but was not the final used. The preprocessing pipeline transforms raw presentation slides into structured passages through four sequential stages. First, shape extraction retrieves text from slide elements, linearizes tables, extracts embedded image text via Tesseract OCR, and pulls speaker notes. Next, text cleaning normalizes Unicode (NFKC), removes noise such as URLs and isolated slide numbers, and maps mathematical logic symbols to readable text. The story assembly stage then hierarchically orders the cleaned content, deduplicates repeated information, and concatenates it into a single continuous narrative. Finally, passage chunking uses the NLTK library to divide this narrative into segments of ≤ 350 tokens, formatting them according to the CoQA standard for downstream model processing. For measurement, the following were used. Answerability (QA-based F1 Score): Checks if questions can be answered by a QA model (Higher scores $\rightarrow$ questions are meaningful, coherent, and relevant). ParaScore / Semantic Similarity Assesses semantic closeness using embeddings, showing if meaning is preserved with persona adaptation.

Question Answer Validation Dataset: The dataset saf_communication_networks_english was used. This dataset has 300 rows with Technical short-answer questions and student-provided responses regarding communication networks. For measurement, accuracy was used.

B. Results

To validate the effectiveness of the proposed framework, each module was evaluated independently using its corresponding benchmark dataset and state-of-the-art models. Performance was quantitatively measured using widely adopted evaluation metrics, enabling a comprehensive assessment of classification accuracy, predictive capability, and overall system reliability. The evaluation criteria employed in this study are presented below.

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

$$\text{Macro F1} = (1/N) \sum F1$$

$$F1 = 2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$$

Table 2. Facial Emotion Recognition Results.

Model	Dataset	Accuracy
GiMeFive	RAF-DB	84.46%
Conv-Cut	FerPlus	68.5%

Table 3. Speech Emotion Recognition Results.

Dataset	PCA	Accuracy	Macro F1
R+T+S+E+C	92.9%	97	0.933

Table 4. Body Language Emotion Recognition Results.

Model	Technique	Accuracy
KNN	Subject dependant	69%
KNN	Subject independant	52.4%
SVM	Subject dependant	69%
SVM	Subject independant	52.4%
ST-TR	Subject dependant	73%
ST-TR	Subject independant	56.3%
MLP	Subject dependant	79.7%
MLP	Subject independant	83.2%
CNN	Subject dependant	89.5%
CNN	Subject independant	75.5%

Table 5. QG Results.

Dataset	CoQA dev set	CoQA dev set	QG
stories	16	100	100
QA pairs	96	800	640
Answerability	0.906	0.856	0.628
Parascore	0.664	0.654	0.600
Bloom Alignment	92.71%	84.75%	85.78%
Anchor Presence	98.6%	97.5%	97.03%

Table 6. QAV Results.

Training Tech	Accuracy
Zero-Shot using DSPy ChainOfThought	72.48%
Zero-Shot using Manual prompt engineering	72.26%

C. User Interface

The Elevare platform is designed to bridge the communication gap across academic and professional environments. The primary user groups and their specific interactions with the system



Fig. 15. The starter page that the user lands in and uploads the presentation video.

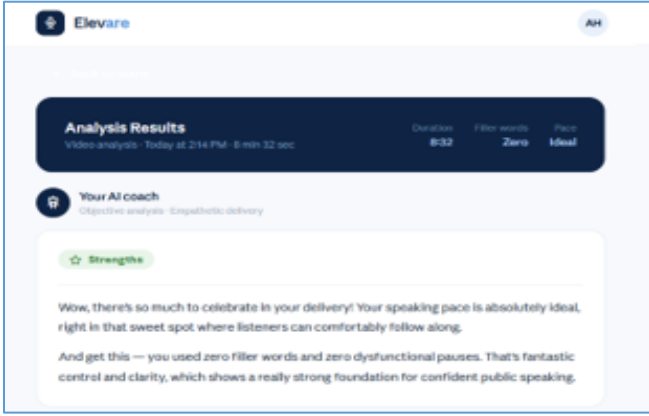


Fig. 16. The page that views the feedback for the user.

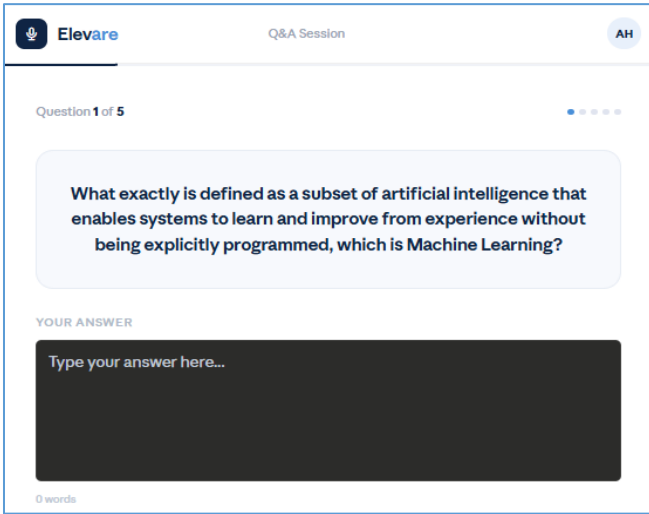


Fig. 17. The page that presents the questions to the user and takes their answers for grading.

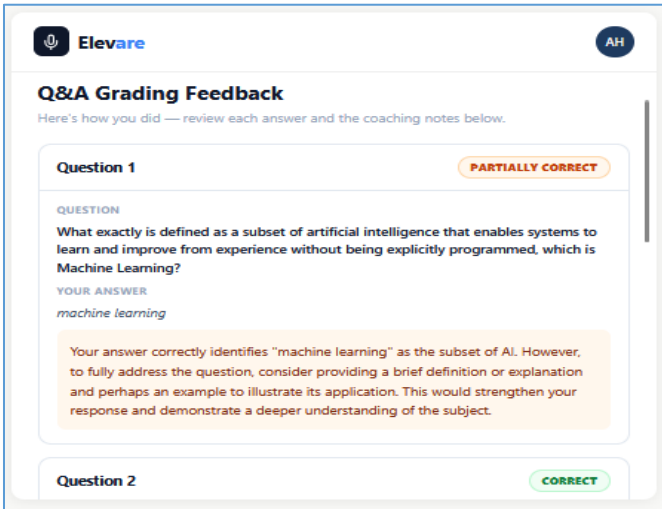


Fig. 18. The page that shows the grading for you user on each question.

V. CONCLUSION AND FUTURE WORK

This paper presented Elevare, an AI-powered multimodal public speaking coach designed to overcome the limitations of conventional presentation assessment systems by providing a holistic, context-aware coaching experience. Unlike existing approaches that primarily focus on isolated speech-related metrics, the proposed framework integrates

speech, facial expressions, body language, presentation understanding, intelligent question generation, answer evaluation, and personalized feedback into a unified architecture. Experimental results demonstrated the effectiveness of the proposed models, achieving 92.90% accuracy for Speech Emotion Recognition, 84.46% for Facial Emotion Recognition, 89.5% for Body Language Emotion Recognition, and 72.48% grading accuracy for presentation answer evaluation. Furthermore, the proposed Dual-Agent Feedback Architecture successfully combines objective performance assessment with personalized coaching while mitigating hallucination and subjective bias in large language models. By integrating multimodal emotion recognition with content-aware presentation analysis, Elevare transforms conventional speech evaluation into an interactive presentation simulator that supports both communication skills and presentation mastery. The proposed framework offers a practical solution for students, researchers, and professionals seeking intelligent, personalized public speaking training and establishes a promising foundation for future AI-assisted coaching systems.

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